Index laws

1

a 2^{21} b 22^{8} c 11^{4} d 7^{5}

e 5^{48} f 15^{4} g 23^{79} h 17^{8}

i 19^{5} j 12^{7} k 13^{8} l 15^{27}

2

a 8 b 7 c 34 d 5

e 3 f 6 g 5 h 38

i 5 j 2

3

a $1\,679\,616$ b $9\,261$ c $1\,024$ d $390\,625$

e $16\,807$ f $4\,096$ g 121 h 729

i 36 j 1 k $1\,728$ l 36

Negative bases

4

a $(-11)^{13}$ b $(-21)^{8}$ c -15^{2} d $(-3)^{7}$

e -12 f $(-30)^{3}$ g 8^{3} h 11^{11}

i $(-7)^{5}$ j $(-6)^{2}$

5

a -7 b -3 c 50 d 2

e 49 f 12 g -2 h -21

i 3 j -55

6

a 256 b 64 c -243 d $-8\,000$

e -27 f $-3\,125$ g -216 h $-1\,977\,326\,743$

i $1\,225$ j $531\,441$ k 64 l $-8\,000$

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Fractional bases

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a $\frac{1}{3^4}$ b $\frac{3^3}{8^3}$ c $\frac{1^8}{4^8}$ d $\frac{5^2}{2^2}$

e $\frac{10^5}{33^5}$ f $\frac{2^6}{35^6}$ g $\frac{5^3}{18^3}$ h $\frac{29^7}{41^7}$

i $\frac{11^9}{13^9}$ j $\frac{20^2}{3^2}$ k $\frac{17^4}{4^4}$ l $\frac{31^5}{50^5}$

8

a 3 b 3 c 2 d 1

e 3 f 16 g 9 h 625

i 2 j 121 k 5 l 5

9

a $\frac{1}{9}$ b $\frac{27}{125}$ c $\frac{1}{32}$ d $\frac{49}{121}$

e $\frac{27}{512}$ f $\frac{144}{169}$ g $\frac{125}{729}$ h $\frac{16}{81}$

i $\frac{4\,096}{15\,625}$ j $\frac{8}{125}$ k $\frac{25}{49}$ l $\frac{1\,000}{2\,744}$

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a No b Yes c No d Yes

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